

LC 207: Course Schedule

Authored by: Arpit Raj, LNMIIT Jaipur

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Goldman Sachs | Flipkart | Swiggy

Problem Overview

- n courses
- [a, b] means a must be studied before b
- **Output:** True/False (T/F) whether all courses can be studied

We can take only those courses whose prereq is 0

→ hence INDEGREE

→ hence KAHN's Algorithm

Example Walkthrough

Graph Representation:

0 → 1

0 → 2

2 → 3

Indegree Table:

Node	Indegree
0	0
1	1
2	1
3	1

Queue Processing:

Now take 0 → push to queue.

Since it's done:

- `indeg[1]--`
- `indeg[2]--`

New Indegrees: 1: 0, 2: 0

→ Next pushed to the queue.

Cycle Detection Edge Case

Example Cycle:

0 → 1

↑ ↓

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Both have indegree 1. Queue stays empty.

0 processed, total 2.

→ **return false**

KAHN TEMPLATE

- S1** build adjacency list
- S2** find indegree
- S3** push all indegree 0 nodes

| C++ Implementation

```

class Solution {
public:
    bool canFinish(int n, vector<vector<int>>& prereq) {

        //basic crux ye he ki
        //agar kisika prereq nhi he to wo course le skte
        //prereq nhi matlb indeg 0
        vector<vector<int>> adj(n);
        vector<int> indeg(n,0);

        for(auto &it:prereq){
            int u=it[0];
            int v=it[1];

            adj[u].push_back(v);
            indeg[v]++;
        }

        queue<int> q;
        for(int i=0;i<n;i++){
            if(indeg[i]==0){
                q.push(i);
            }
        }

        int took=0;
        while(!q.empty()){
            int curr=q.front();
            q.pop();
            took++;

            for(auto &it:adj[curr]){
                indeg[it]--; //prereq course padhne ke baad future course
                padhne ke closer
                if(indeg[it]==0){
                    q.push(it);
                }
            }
        }

        if(took==n) return true;
        else return false;
    }
};

```